

BIOS 343: Animal Physiology (3 crd)

Course Guide and Syllabus – Fall 2023

Welcome to Biology 343!

Description and Prerequisite Statement: How the human body works including digestive, neuromuscular, sensory, respiratory, excretory, endocrine, and cardiovascular systems using examples across animal species and human case studies. Course Information: Previously listed as BIOS 240. Credit is not given for [BIOS 343](#) if student has credit in BIOS 240. Prerequisite(s): BIOS 100 or [BIOS 110](#); and BIOS 101 or [BIOS 120](#). Recommended background: [BIOS 222](#).

INSTRUCTOR INFORMATION

Alex Daemicke

Drop-In Hours: T 11:30 – 12:30 pm (in office), Th 11 am-12 pm (review - Zoom), and by appointment
Office: SES 3348
Email: adaemick@uic.edu

Your TA Information

Alex Kontos: akonto3@uic.edu
Drop-In Hours: M 3-5 pm (Zoom)

Elissa Schaffner: eschaf4@uic.edu
Drop-In Hours: F 9-11 am (Zoom)



COURSE INFORMATION

Time (On Campus)

Lecture: TR 8 – 9:15 am
Behavioral Sciences Building | Room 250

My Teaching Philosophy

When reflecting on the teachers that have made the most profound impact on my life, I cannot help but identify how they assumed the role of a guide, helping me acquire a better understanding of the didactic material, as well as myself as a learner. Thus, I hope to do the same by helping you make connections, both within yourselves and with the material, and supporting the acquisition of knowledge as you move closer towards achieving your goals. I recognize that the educational experience does not occur within a vacuum, but rather it can be heavily influenced by factors such as prior knowledge, past lived experiences, culture, belief system, as well as physical and mental health.

While it is always my goal as an instructor to have students finish the semester with a deeper understanding and appreciation for the course material, something I argue to be equally

important is the role I can play in facilitating a change in the way you approach your learning and ability to think critically. To do so, rather than asking you to regurgitate factual information, my activities and assessments will encourage you to reflect on your learning and apply your knowledge to novel scenarios in ways that actively engage the material.



Course Learning Goals

This course serves to fulfill the following goals, concepts, and skills.

Overarching Course Goals

- Improve knowledge of common biological terms, facts and principles through quizzes, class sessions and written assignments.
- Encourage construction of ideas and connections between concepts to make sense of the ideas rather than simply memorizing isolated facts.
- Strengthen understanding of the "scientific approach" by its application in-class discussions, written responses to problems and interpretations of content presented during class.
- Provide an adequate foundation in physiology content that will enable students to build upon in higher level courses.

Regular Course Objectives

Concepts

- Evolution: Students will discover the patterns, processes, and relationships that lead to the diversity within animal life.
- Structure and Function: Students will evaluate how form determines the function of all levels of organization.
- Information Flow: Students will analyze how all levels of biological organization depend on specific interactions and information transfer.
- Systems: Students will relate the dynamic interactions of components at one level of biological organization to the functional properties that emerge at higher organizational levels.

Course Skills

- Students will apply the process of animal science to real life situations
- Students will individually and collaboratively model scientific thinking
- Students will relate biology to other sciences (e.g., chemistry, physics)

*All the above concepts and skills are in alignment with Vision and Change in Undergraduate Biology Education (2009)

COURSE EXPECTATIONS

Method of Instruction

This is a synchronous in-person course. Every T/Th 8-9:15 am we will be meeting in Behavioral Sciences Building | Room 250.

Scheduled topics and the days that we plan to review them are outlined in the course schedule (pg. 7-8). It is strongly encouraged that you attend these lecture sessions to best prepare for assignments and quizzes. To promote active learning and critical thinking development, activities presented during class sessions, assignments and assessments will be integrated throughout the semester.

- **Use of Computers During Lectures: A Note on Taking Notes** - As you will discover, all science is based on evidence, and evidence indicates that laptop note-taking is less effective than longhand-note taking for learning (*Mueller and Oppenheimer, 2014*). Further, the temptation to check Facebook, your email, Twitter, etc., is strong when you have your computer open, and is distracting to those around you. Consequently, take notes on paper during the live sessions or when reviewing the recording rather than on the computer. This consolidation process is critical to learning, as it helps your brain make sense of the information you take in, make connections with other information, and file the information away in the correct part of your long-term memory.

Blackboard Course Site

Students are expected to log into the course site regularly to learn about any developments related to the course, upload assignments, and communicate with classmates. For all technical questions about Blackboard, email the Learning Technology Solutions team at LTS@uic.edu.

Drop-In Hours

One of the best aspects of being an instructor is learning more about my students, where you come from, where you are at, and where you want to go! Consequently, drop-in hours are not just for talking about the course content: it's a place to talk about anything that impacts your academic success, both positive and negative. Drop-in hours are reserved for you – please take advantage of them! These will take place in my office (SES 3264).

- **Answering emails:** Students who have questions or want to make appointments should contact me by email. I will endeavor to answer emails within 24-48 hours. I do not check emails after 8:30 pm during the work week and Saturdays. If an email is sent late Friday or on the weekend, I will attempt to answer such emails within 24 hours of the next business day. Students are responsible for all information instructors send to your UIC email and Blackboard accounts.
- **Do-not-reply emails:** Please do not respond to emails generated by Blackboard that go out to the entire



class. If you wish to follow up create a new email thread and only include the individual(s) you seek to communicate with.

Review Sessions

Once a week I will host a one-hour review session on Thursdays 11-12 pm via Zoom, during which we will review content that will be assessed on upcoming quizzes by completing the live lecture outlines found in our weekly folders. These are not mandatory and are recorded.

Recommended Course Materials

Hill et al., *Animal Physiology*, fourth edition (other editions are fine)

- Free online .pdf is uploaded to Bb!

Technology Requirements

- Computer Requirements: You will need regular access to a personal computer that runs on a broadband Internet connection.
- Blackboard Learning Management System: For all technical questions about Blackboard, email ACCC-Learning Technology Solutions at LTS@uic.edu
- Zoom: We will be using Zoom to conduct our review sessions. Please make sure to download the program from: <https://zoom.us/download>

Student Resources and Support

- Outlines: Under each weekly content folder, you will find a live session outline. Many students like to take notes while listening to the lectures, which is great! These documents help with this task and serve as a handy tool to help you prepare for our assessments.
- Study Skills Resources: Suggestions for how to improve study strategies found under the *Stop and Think Exercises* tab.
- Syllabus: This document is certainly long but it is so for a reason! It is your one stop shop resource about questions related to the structure of the course. Images included credited to LyonRoadArt (Etsy Shop).
- Course materials and assignments can be complex and challenging, but they are crucial to your intellectual and personal growth and development. There are times you may need extra help. Students who attend class consistently, complete all assignments, thoughtfully engage with feedback on work, develop good study strategies, visit the tutoring center, and contact faculty when struggling can develop a thorough understanding of the course material and ultimately succeed in the course!

Evaluation

You will be asked to fill out an evaluation at the end of the course. However, you do not have to wait until the end of the year to provide an assessment of the course design and implementation relative to your own learning. The format of this course provides some flexibility, and I would like to accommodate student needs and interests as much as feasible. Please let me know if there are any reasonable changes that could be made to improve your learning and enjoyment; suggestions, comments, and constructive criticism are encouraged at any time.

Copyright Infringement

The posting of any course material on websites such as Chegg or any similar websites without an instructor's permission is a violation of copyright ownership. This violates rules of academic integrity. Thus, will result in the submission of an Academic Integrity Incident Report form and the associated disciplinary action as outlined in UIC's academic integrity policy. I am grateful for your cooperation in honoring this important request.

Fabrication (pg. 14): https://dos.uic.edu/wp-content/uploads/sites/262/2020/01/DOS_Student-Disciplinary-Policy-2020.pdf

Academic Integrity Incident Report Form:

https://cm.maxient.com/reportingform.php?UnivofIllinoisChicago&layout_id=10

ASSIGNMENTS & ASSESSMENTS

Quizzes

We will have a quiz during the last lecture session of each unit and during finals week. Each quiz will be a combination of multiple choice and T/F making up a total of 50 points. Quizzes will include questions drawing on material from lectures, activities, and additional readings from the entire unit! If a quiz is not taken, the score for that quiz will be counted as a zero. One quiz score will be dropped at the end of the semester. These assessments will be worth a total of 200 points $((50 \times 5) - 50 = 200)$.

Knowledge Checks

The second week of each unit, we will have a short 30-point Knowledge Check to encourage that everyone keeps up on the material. Unlike the Quizzes, Knowledge Checks will be available on Friday morning at 8 am and close at 11:59 pm the same evening on Blackboard in their respective weekly folders. These assessments will be worth a total of 90 points as your lowest Knowledge Check will be dropped. Once you start you will have 35 minutes to complete it in one sitting. The Knowledge Check of a unit will assess the first and second week of the unit's content.

Any lecture that I start halfway through the Thursday session WILL NOT make an appearance on the Knowledge Check. I might do this occasionally to get ahead for the next week.

Below you will find an example Quiz and Knowledge Check schedule.

Unit 1	Assessment
Week 1	No assessment
Week 2	Knowledge check covering Weeks 1 & 2 content
Week 3	No assessment
Week 4	Quiz 1 covering mainly content from Weeks 1-4

NO MAKE-UP ASSESSMENTS without legitimate excuse (e.g., death in the family, religious holiday).



Unit Packets

There will be a packet of worksheets for each unit, totaling 40 points each. These are designed to help students become comfortable with and capable of mastering the concepts covered in the course. Worksheets will include a bit of research, sequencing of events and clinical application activities. Please **type all answers** and submit them in .pdf or .docx form! Unit packets are due by 11:59 pm on the final Sunday of each unit except for the Unit 4 Packet which will be due on the final Wednesday (11/29) of the semester. (160 pts total)

*** If you are a mac- user, do not submit as a .pages file. Convert to .docx or a .pdf so that your work can be rendered by the BB doc-viewer and graded. Blank or corrupted files will not be accepted.

Animal Comparison Assignment

Twice during the semester, I will ask you and your group members will complete a 2-3 page write up comparing and contrasting the physiology of one of the three organisms assigned to you with that of a human, focusing on one organ system that we have discussed in that half of the semester. For the second portion of this assignment, you and your group members will create your own clinical application problem. This will integrate information about the organ system and animal you discussed in your write up. Along with your problem you this activity serves to promote collaboration amongst students and challenge critical thinking skills. More information on this project will be provided within the first couple weeks of class. (80 points total)

Stop and Think Exercises

Metacognition is a component of self-regulated learning that is often defined as, "thinking about one's thinking." Research finds that students that have stronger metacognitive skills perform better in science related courses. I am a firm believer in the power of metacognition in the learning process and want to help all of you develop strong reflective learning skills that will serve you well even after we have concluded our course. Several activities will be assigned throughout the semester to help you develop such skills. These exercises are due by 11:59 pm on Sunday of the week when they are assigned. (20 points total)

Extra Credit ☺

For each unit, you may choose to submit one of the two extra credit assignments each unit. **Considering this, any requests for grade 'bumps' will not be honored.** Extra credit is due by 11:59 pm on the final Sunday of each unit.

- **Quiz Creation:** Writing benefits the development of critical thinking skills as it requires you to make your ideas explicit, as well as evaluate and choose among tools necessary for effective communication. Students exhibit larger gains in critical thinking

skills when participating in biology courses that involve inquiry, writing, research, and analysis compared with students in traditional courses. This semester you will have any opportunity to engage in such a generative activity by way of creating quizzes! By generating your own quiz questions and supportive answer keys, you will practice skills associated with advanced critical thinking skills (2.5 pts).

- **Science in the News:** Additionally, in efforts to connect our course to everyday life, you will receive extra credit if you find and read a current article (within the past 3 years) related to the topics covered in lecture. These can be short articles that appear in such venues as the [New York Times](#), Science News, or any research journal. Whatever the source may be, it must be reputable and [cited correctly](#). After reading the article you will need to submit a summary of what you learned and state what questions might have arisen as a result of reading the article (2.5 pts).

GRADING POLICY AND BREAKDOWN

Your grade reflects your mastery of the material. As in any course, getting a good grade demands hard work, clear thinking, and good organizational skills. Grades will not be assigned on a curve. A final course grade will be determined by examining the total number of points earned during the semester based on the following assignment point values:

Course Assessments	Total Points	~% of Final Grade
Quizzes (4x50 pts)	200	36.4%
Knowledge Check (3x30 pts)	90	16.4%
Unit Packets (4x40 pts)	160	29.1%
Animal Comparison Assignment (2x40 pts)	80	14.5%
Stop and Think Activities (4x5 pts)	20	3.6%

Total ----- 550 pts

Extra Credit Opportunities	MAX 12.5 PTS
Syllabus Quiz	2.5
Quiz Creation or Science in the News	2.5/unit

The following scale will be used to generate your final grade:

A: 495-550 (90-100%) **B:** 440-494.99 (80-89.99%) **C:** 385-439.99 (70-79.99%) **D:** 330-384.99 (60-69.99%) **F:** 0-329.99 (>59.99%)

Grades will be posted on the UIC Blackboard course site. Make sure to check the accuracy of your grades in the grade book on a regular basis.

- **Late Policy:** Due dates are very important, and this class will adhere to posted due dates. If there are extenuating circumstances preventing you from completing your coursework (e.g., death in the family) please reach out to be as soon as possible. Also, I understand technical issues sometimes crop up, so I am willing to be flexible with these issues so long as there is prompt communication. The official late policy, though, is that you will receive a 10% deduction for every day the assignment is late.



Course Schedule – Lecture

Week	Start Date	Topic	Assignments Sunday 11:59pm	Assessment KC via Bb Quiz in person	Textbook
1 UNIT 1	Aug 21	T: Intro to Course/Fundamentals of Physiology Th: Fundamentals of Physiology			1, 2
2	Aug 28	T: Membrane Transport Th: Transport and Reception; <i>Project Overview and Check-In (Last 15 mins)</i>	S&T: Getting to Know You	Knowledge Check 1	5
3	Sept 4	T: Neural Communication* Th: Neural Comm/Synaptic Transmission	S&T: Time Management Exercises		12, 13
4	Sept 11	T: NS Organization and Autonomic NS Th: Start Phototransduction but not on Quiz 1	Unit 1 Packet Unit 1 ExCrd		15
5 UNIT 2	Sept 18	T: Quiz 1 Th: Phototransduction/Circadian Rhythm Regulation	S&T: Post Unit 1 Check-In	Quiz 1	14, 15
6	Sept 25	T: Muscle Physiology Th: Muscle Physiology		Knowledge Check 2	19, 20
7	Oct 2	T: Respiration Th: Respiration*	Animal Comparison 1		22-24

8	Oct 9	T: Bone Physiology Th: Start Circulation but not on Quiz 2	Unit 2 Packet Unit 2 ExCrd		25
9 UNIT 3	Oct 16	T: Quiz 2 Th: Circulation	S&T: Midsemester Reflection	Quiz 2	29
10	Oct 23	T: Digestion Th: Digestion/Osmoregulation		Knowledge Check 3* *Osmoreg not on KC	6
11	Oct 30	T: Osmoregulation Th: Start Hormonal Communication but not on Quiz 3	Unit 3 Packet Unit 3 ExCrd		7
12 UNIT 4	Nov 6	T: Quiz 3 Th: Hormonal Communication		Quiz 3	16
13	Nov 13	T: Reproduction Th: Reproduction	Animal Comparison 2	Knowledge Check 4	17
14	Nov 20	T: Development Th: Thanksgiving! 🦃			
15	Nov 27	T: Student Choice Th: Quiz 4	Unit 4 Packet* *due Wed, Nov 29th 11:59 pm Unit 4 ExCrd	Quiz 4	
16	Dec 4	Optional Final Quiz (DAY/TIME TBD)			

Disclaimer

This syllabus is intended to give the student guidance on what may be covered during the semester and will be followed as closely as possible. However, as the instructor, I reserve the right to modify, supplement, and make changes as course needs arise. I will communicate such changes in advance through in-class announcements and in writing via Blackboard Announcements.

Our Weekly Schedule

Monday	Tuesday	Wednesday	Thursday	Friday	Sat/Sun
Weekly news brief	Live session 8 am ADH: Drop-in 11:30-12:30 pm	Midweek minutes	Live session 8 am ADH: Review 11-12 pm	KCs administered through Bb 8am – 11:59pm	Assignments (when applicable) due by Sunday 11:59 pm (except Unit 4 Packet)

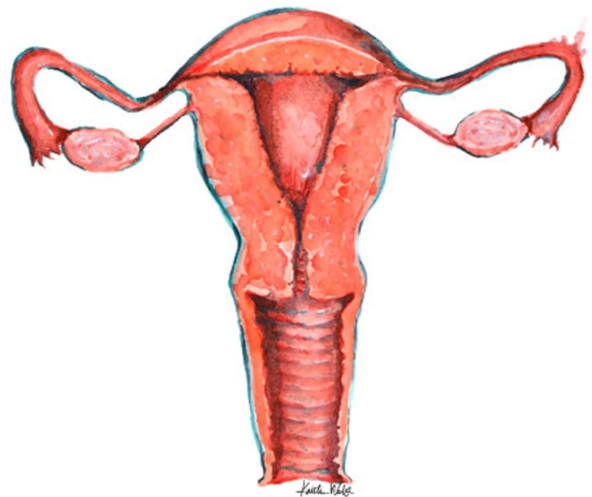
Classroom Policies/Procedures

Student Conduct

- Each student is responsible for adhering to the Code of Student Conduct as stated in the college catalog. Students are expected to demonstrate mature and appropriate behavior when interacting with each other, with the professor, and with the teaching assistants. Any student being disruptive during class, discussion, laboratory, or during an office hour will be asked to leave.
- Academic Integrity:** As a student and member of the UIC community, you are expected to adhere to the [Community Standards](#) of [academic integrity](#), accountability, and respect. Please review the [UIC Student Disciplinary Policy](#) for additional information.

Attendance

- Please email me if you face an unexpected situation that may impede your attendance, participation in required class and assessment sessions, or timely completion of assignments.
- It is encouraged that students attend live classes. The learning activities, lectures, and experiences are difficult and are often not able to be replicated in a recording. Attendance has a direct impact on meeting the course outcomes and objectives; therefore, students should attempt to have a few absences as possible.
- Withdrawal: Friday, September 1st is the last day to complete late registration, add a course(s) or make section changes, drop individual courses via Student Self-Service without receiving W (Withdrawal) grade on academic record. This is the last day to



Web Drop courses via XE Registration and received 100% cancellation of tuition and fees.

- Please email me if you aren't able to attend class, particularly if we have a quiz, as a result of circumstances that are beyond your control such as testing positive for COVID-19, illness, emergency or other highly sensitive matters.

Cheating and Plagiarism

- Any student caught cheating on a quiz, packet or any other assignment may be immediately dismissed from the course and given a final grade of "F" regardless of the student's total points in the course.

Accommodations

- **Disability Accommodation Procedures:** UIC is committed to full inclusion and participation of people with disabilities in all aspects of university life. If you face or anticipate disability-related barriers while at UIC, please connect with the Disability Resource Center (DRC) at drc.uic.edu, via email at drc@uic.edu, or call (312) 413-2183 to create a plan for reasonable accommodations. To receive accommodations, you will need to disclose the disability to the DRC, complete an interactive registration process with the DRC, and provide me with a Letter of Accommodation (LOA). Upon receipt of an LOA, I will gladly work with you and the DRC to implement approved accommodations.
- **Religious Accommodations:** Following [campus policy](#), if you wish to observe religious holidays, you must notify me by the tenth day of the semester. If the religious holiday is observed on or before the tenth day of the semester, you must notify me at least five days before you will be absent. Please submit [this form](#) by email with the subject heading: **"YOUR NAME: Requesting Religious Accommodation."**

Letters of Recommendation Policy

- I am happy to write letters for students who exhibit investment in and enthusiasm for my courses. The best letters come about when I get to know you so if this is something you'd like from me in the future please introduce yourself and visit me during drop-in hours!
- All LOR requests must be made by **March 1st** before the cycle in which you intend to apply.

Classroom Environment

- **Inclusive Community:** UIC values diversity and inclusion. Regardless of age, disability, ethnicity, race, gender, gender identity, sexual orientation, socioeconomic status, geographic background, religion, political ideology, language, or culture, we expect all members of this class to contribute to a respectful, welcoming, and inclusive environment for every other member of our class. If aspects of this course result in barriers to your inclusion, engagement, accurate assessment, or achievement, please notify me as soon as possible.
- **Name and Pronoun Use:** If your name does not match the name on my class roster, please let me know as soon as possible. My pronouns are [\[she/her; he/him; they/them\]](#). I welcome your pronouns if you would like to share them with me. For

more information about pronouns, see this page: <https://www.mypronouns.org/what-and-why>.

Community Agreement/Classroom Conduct Policy

- Be present by silencing cell phones and removing yourself from other distractions.
- Be respectful of the learning space and community. For example, no side conversations or unnecessary disruptions.
- Use preferred names and gender pronouns.
- Assume goodwill in all interactions, even in disagreement.
- Facilitate dialogue and value the free and safe exchange of ideas.
- Try not to make assumptions, have an open mind, seek to understand, and not judge.
- Approach discussion, challenges, and different perspectives as an opportunity to “think out loud,” learn something new, and understand the concepts or experiences that guide other people’s thinking.
- Debate the concepts, not the person.
- Be gracious and open to change when your ideas, arguments, or positions do not work or are proven wrong.
- Be willing to work together and share helpful study strategies.
- Be mindful of one another’s privacy, and do not invite outsiders into our classroom.

- **Content Notices and Trigger Warnings:** Our classroom provides an open space for a critical and civil exchange of ideas, inclusive of a variety of perspectives and positions. Some readings and other content may expose you to ideas, subjects, or views that may challenge you, cause you discomfort, or recall past negative experiences or traumas. I intend to discuss all subjects with dignity and humanity, as well as with rigor and respect for scholarly inquiry. If you would like me to be aware of a specific topic of concern, please email or visit my drop-in hours.

Resources: Academic Success, Wellness, and Safety

We all need the help and the support of our UIC community. Please visit my **drop-in hours** for course consultation and other academic or research topics. For additional assistance, please contact your assigned college advisor and visit the support services available to all UIC students.

Academic Success

- UIC [Tutoring Resources](#)
- [UIC Library](#) and [UIC Library Research Guides](#).
- [Offices](#) supporting the UIC Undergraduate Experience and Academic Programs.
- [Student Guide for Information Technology](#)
- [First-at-LAS](#) Academic Success Program, focusing on LAS first-generation students.

Wellness

- **Counseling Services:** You may seek free and confidential services from the Counseling Center at <https://counseling.uic.edu/>.
- Access [U&I Care Program](#) for assistance with personal hardships.

- **Campus Advocacy Network:** Under Title IX, you have the right to an education that is free from any form of gender-based violence or discrimination. To make a report, email TitleIX@uic.edu. For more information or confidential victim services and advocacy, visit UIC's Campus Advocacy Network at <http://can.uic.edu/>.

Safety

- [UIC Safe App](#)—PLEASE DOWNLOAD FOR YOUR SAFETY!
- [UIC Safety Tips and Resources](#)
- [Night Ride](#)
- [Emergency Communications](#): By dialing 5-5555 from a campus phone, you can summon the Police or Fire for any on-campus emergency. You may also set up the complete number, (312) 355-5555, on speed dial on your cell phone.